

Sparse Spectral LoRA: Routed Experts for Medical VLMs

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Abstract

Large vision-language models (VLMs) excel on general benchmarks but often lack robustness in medical imaging, where heterogeneous supervision induces cross-dataset interference and sensitivity to data regime (i.e., how the supervisory signals are mixed). In realistic clinical workflows, data and tasks arrive sequentially, so naive continual training further leads to catastrophic forgetting. To address these challenges, we propose *MedQwen*, a parameter-efficient medical VLM that couples a spectrally routed Mixture-of-Experts (MoE) with a theoretically grounded scaling rule that aligns low-rank updates with a full-rank, fully fine-tuned MoE, without changing the base architecture. Concretely, we initialize each expert from non-overlapping singular value decomposition (SVD) segments of the pretrained weight and introduce a residual compensation and scaling scheme to enable stable expert specialization and consistent routing under distribution shift. Across 23 medical datasets covering visual question answering, report generation, radiology classification, and hallucination mitigation, *MedQwen* achieves strong, reliable performance: it approaches full fine-tuning on zero-shot classification with $339\times$ fewer trainable parameters, and reduces sequential forgetting to $\sim 5\%$ where strong baselines degrade by $>20\text{--}50\%$.

1. Introduction

Vision-Language Models (VLMs) have demonstrated strong generalization across open-vocabulary tasks by jointly learning from paired images and text [39, 52, 58, 62]. Despite impressive general-domain results, off-the-shelf VLMs require adaptation to medical tasks [2, 18, 60, 69]. Specialized medical VLMs, including Med-Flamingo [58], HealthGPT [44], and LLaVA-Med [39], have shown strong performance in medical visual question answering (VQA) and related tasks. However, these models often lack the versatility needed for both discriminative and generative clinical tasks. Moreover, their reliance on large backbones (e.g., the 7B-parameter LLaMA in LLaVA-Med) incurs heavy training and inference costs. Parameter-Efficient Fine-Tuning (PEFT) methods such as Low-Rank Adaptation (LoRA) [24] mitigate these challenges by updating only lightweight low-rank ma-

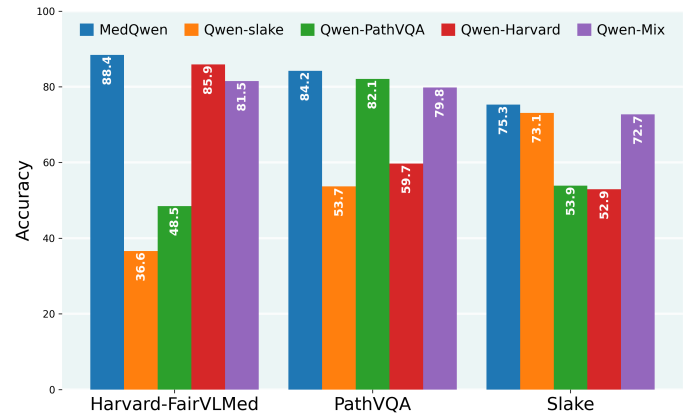


Figure 1. Model performances on three benchmarks when trained with different data configurations.

trices while keeping pre-trained weights frozen, enabling parameter-efficient domain adaptation of medical VLMs [50, 88], but this alone has not resolved the reliability and cost constraints that are important in practice.

Our investigations reveal that current VLMs fine-tuned using standard LoRA exhibit high sensitivity to the *training data regime* [6, 40]: models tuned on one medical dataset (e.g., Qwen-Slake in Fig. 1) often regress on others, and naively mixing heterogeneous sources introduces cross-dataset interference (e.g., Qwen-Mix in Fig. 1). This data-regime instability limits scalability across modalities, anatomies, and clinical tasks. Moreover, while tuning with broader data often reduces hallucinations [17, 48, 70], it could increase *catastrophic forgetting* when tasks arrive sequentially, which is pervasive in clinical settings.

To address these limitations, we introduce *MedQwen*, a routed, *SVD-structured LoRA Mixture of Experts (MoE)* that turns the singular-value spectrum of pre-trained weights to a set of non-overlapping spectral priors. Each expert is initialized from a distinct SVD segment, and a lightweight router activates only the few experts whose priors are most relevant for the current input. This design encourages specialization without discarding useful pre-training structure and directly targets cross-dataset interference. To stabilize optimization, we derive a theoretical scaling scheme that *aligns low-rank updates with the gradient geometry of a fully fine-tuned MoE*, improving convergence without changing the base architecture or optimizer.

Extensive experiments on 23 medical datasets spanning

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VQA, report generation, and classification validate the approach. MedQwen reaches SOTA accuracy among PEFT methods while preserving efficiency, approaches full fine-tuning quality, and exhibits robustness to sequential training: on a Harvard-FairVLMed to PathVQA protocol, a standard LoRA model loses over 50% of its original accuracy, MoE-LoRA loses over 20%, whereas MedQwen’s accuracy drops by only 5%. We further analyze expert sparsity, rank, and routing, demonstrating favorable compute–performance trade-offs within a single-GPU budget. Our contributions are:

- **Sparse spectral LoRA:** We introduce an SVD-structured MoE that partitions pre-trained weights into non-overlapping spectral segments and routes inputs to the most relevant low-rank experts, reducing cross-dataset interference and preserving useful priors.
- **Optimization alignment with scaling:** We establish conditions under which LoRA-MoE matches full-rank MoE dynamics, provide a residual-matching initialization, and derive a theoretical scaling factor that preserves gradient geometry under low rank.
- **Unified, efficient medical VLM:** Without architectural changes, MedQwen achieves SOTA results across a wide set of medical VQA, report generation, and classification tasks while remaining efficient.
- **Comprehensive analysis:** We ablate expert count, activation sparsity, rank, and scaling, and study convergence and compute trade-offs.

2. Related Work

VLMs in Medicine. Medical VLMs (Med-VLMs) have yielded excellent capabilities in image interpretation and VQA [49, 59]. XrayGPT [67] integrates a specialized visual encoder (MedClip) [79] with a fine-tuned VLM, employing a straightforward linear transformation layer to achieve alignment between visual and textual modalities. LLaVA-Med [39] further refines visual-textual alignment in medical contexts by curating high-quality image-text pairs from PubMed publications and synthetic VQA datasets. BiomedGPT [56] utilizes a BERT-style encoder and a GPT-style decoder architecture, trained on multimodal datasets, outperforming much larger commercial models like Med-PaLM [64]. HuatuoGPT-Vision [5] scale supervision with PubMedVision dataset (1.3 million medical samples) to improve adaptability across tasks. Despite this progress, most Med-VLMs are optimized for conversational VQA/report generation and rely on PEFT, and broad versatility across discriminative and generative clinical tasks typically requires additional task-specific heads or training.

Mixture of Experts. MoE models expand model capacity by activating only a subset of experts per token, enabling a sub-linear increase in computational cost in Transformers while retaining large representation power [8, 29, 63, 92]. Recent MoEs differ in their expert selection and routing strategies. LLaVA-MoE [6] routes tokens to domain-specific experts within Transformer layers, effectively reducing data interference. Other MoE-based methods improve domain adaptation and lifelong learning, such as MoRAL [87], LoRAMoE [12],

and PESC [84], while MoE-LoRA [54] and MoCLE [15] allocate or activate task-specific parameters based on layer or instruction clusters. Our work builds on this line by mixing LoRA experts across diverse medical datasets, highlighting MoE’s potential to mitigate cross-dataset conflicts in multi-modal medical settings.

Parameter-Efficient Fine-Tuning. Given the large size of foundation models, recent research has focused on developing PEFT methods [9, 24, 38, 42], which reduce fine-tuning costs by updating only a small subset of parameters. PEFT approaches can be grouped into three main directions: (1) Additive Rank/Scale: methods such as AdaLoRA [94] and rsLoRA [73] dynamically allocate parameter budgets or adjust scaling factors; (2) Architectural Improvements: DoRA [51] decomposes weights into magnitude and direction, while RoseLoRA [73] applies sparse low-rank adaptation; and (3) Optimization and Initialization: including PiSSA [57], MiLoRA [75], LoRA-Pro [81], and LoRA-GA [76], which refine singular components or align low-rank gradients with full fine-tuning. Despite these advances, a notable performance gap often remains between PEFT methods and full fine-tuning.

SVD-based VLM Adaptation. SVD is a classical tool for compression and latent semantic analysis [21, 22, 66], and has recently emerged as a promising approach for PEFT in VLMs [36, 78, 89]. AdaLoRA [93] and SARA [16] employ SVD to identify suitable ranks, improving parameter efficiency. PiSSA [57] and MiLoRA [74] use SVD for LoRA initialization; PiSSA fine-tunes dominant components for faster convergence, while MiLoRA targets minor components to enhance task-specific adaptability. In contrast, our approach (1) partitions spectral components into non-overlapping experts, (2) adaptively selects relevant priors via routing, and (3) corrects weight misalignment and gradient dynamics through theoretically grounded scaling.

3. Background and Motivation

3.1. SVD Initialization Overview

Initializing LoRA matrices with SVD is a common strategy to keep key structure from pre-trained weights [53, 57, 71, 95]. Existing methods differ mainly in which spectral components they update: PiSSA [57] fine-tunes the dominant singular directions, while MiLoRA [72] targets the minor ones to improve task specificity.

To establish a unified theoretical framework that links SVD-based initialization methods with fine-tuning (FT), let the pre-trained weight matrix be $W^{(0)} \in \mathbb{R}^{m \times n}$, represented by its SVD $W^{(0)} = USV^\top$. Assuming $h = \min(m, n)$ and a target LoRA rank r , we proceed to decompose $W^{(0)}$ into rank- r blocks as follows:

$$W^{(0)} = \sum_{i=0}^l U_i S_i V_i^\top, \quad (1)$$

Here, $l = \frac{h}{r} - 1$, and i refers to the segment $[i \cdot r : (i+1) \cdot r]$. For each segment, the submatrices are defined as $U_i = U_{[:, i \cdot r : (i+1) \cdot r]} \in \mathbb{R}^{m \times r}$, $S_i = S_{[i \cdot r : (i+1) \cdot r, i \cdot r : (i+1) \cdot r]} \in \mathbb{R}^{r \times r}$,

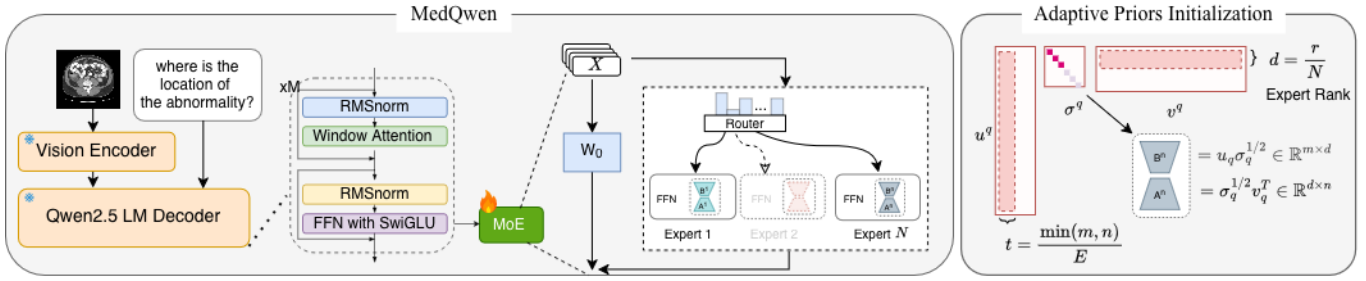


Figure 2. An overview of the proposed MedQwen approach.

and $V_i = V_{[:,i:r:(i+1) \cdot r]} \in \mathbb{R}^{n \times r}$. The subsequent FT procedures in different relevant techniques are expressed as:

$$\begin{aligned}
 \text{Full FT} &: U_0 S_0 V_0^\top + U_1 S_1 V_1^\top + \dots + U_l S_l V_l^\top \\
 \text{MiLoRA} &: (U_0 S_0 V_0^\top + \dots + U_{l-1} S_{l-1} V_{l-1}^\top)^* + U_l S_l V_l^\top \\
 \text{PiSSA} &: U_0 S_0 V_0^\top + (U_1 S_1 V_1^\top + \dots + U_l S_l V_l^\top)^* \\
 \text{KaSA} &: (U_0 S_0 V_0^\top + \dots + U_{l-1} S_{l-1} V_{l-1}^\top)^* + U^r S^r V^{r\top}
 \end{aligned} \quad (2)$$

where $(\cdot)^*$ denotes frozen components. The trainable LoRA parameters are:

$$B = U_i S_i^{1/2} \in \mathbb{R}^{m \times r}, \quad A = S_i^{1/2} V_i^\top \in \mathbb{R}^{r \times n}. \quad (3)$$

We observe that PiSSA [57] freezes the minor singular values and fine-tunes only the components $U_0 S_0 V_0^\top$ with the largest norms, thereby achieving an optimal approximation to $W^{(0)}$. In contrast, MiLoRA [75] and KaSA [71] retain segments $0 \sim (l-1)$ as preserved pre-trained knowledge, while KaSA regards the minor component $U_l S_l V_l^\top$ as noise and replaces it with a new random component $U^r S^r V^{r\top}$. Empirically, PiSSA converges faster by emphasizing the principal singular values, whereas MiLoRA and KaSA maintain more pre-trained knowledge, leading to improved final performance. This observation highlights a key trade-off between focusing on principal components and emphasizing minor components.

According to [13], the performance of FT varies across different singular value segments and depends on the dataset; for example, $x = l$ yields superior performance on the Slake dataset, whereas $x = 0$ performs better on the PathVQA dataset. Additionally, the intermediate segments contribute significantly to model performance. When $r = 128$, the best results are typically achieved in the middle range of segments. These results indicate that each singular value segment carries task-dependent information, thereby motivating an adaptive mechanism that enables the model to automatically select relevant segments during optimization while preserving the intrinsic structure of the pre-trained matrix.

3.2. Scaling Factor Overview

In LoRA, the parameterization $W = W^{(0)} + sBA$ is widely adopted, yet the role of the scaling factor s is often under-analyzed. Biderman et al. [3] suggest setting $s = 2$, the SVD-based approach of PiSSA [57] empirically eliminates the dependency of sBA on s by dividing A and B by $\sqrt{\frac{1}{s}}$, and

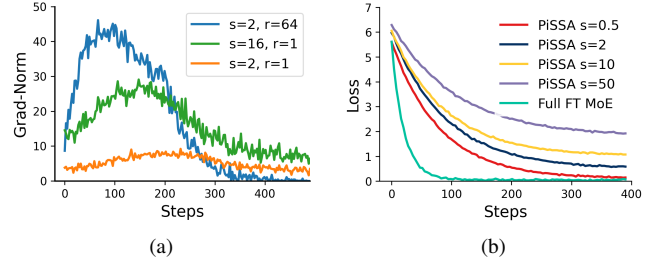


Figure 3. SVD initialization vs. scaling s and rank r .

Tian et al. [68] report that employing larger scaling factors in LoRA-MoE architectures improves performance.

To analyze this effect, as depicted in Fig. 3b, we vary s in the SVD-based LoRA with a fixed rank and observe that s continues to influence the convergence rate. To capture this effect quantitatively, we define:

$$\tilde{W} = W + sBA, \quad \tilde{g} = \frac{\partial L}{\partial \tilde{W}}, \quad (4)$$

where $\tilde{W}$ and $\tilde{g}$ denote the equivalent weight and gradient, respectively (see Fig. 4). Also, L denotes the loss function. At the optimization step t , letting g_t be the full FT gradient and A, B denote low-rank factors, the equivalent gradient is:

$$\tilde{g}_t = s^2 \left(B_t B_t^\top g_t + g_t A_t^\top A_t \right). \quad (5)$$

For SVD-based initialization, we obtain:

$$\tilde{W} \propto sBA = s \left(\frac{1}{s} U_r S_r V_r^\top \right) = U_r S_r V_r^\top, \quad (6)$$

$$\begin{aligned}
 \tilde{g} &= s^2 \left(\frac{1}{s} U_r S_r U_r^\top g + \frac{1}{s} g V_r S_r V_r^\top \right) \\
 &= s \left(U_r S_r U_r^\top g + g V_r S_r V_r^\top \right).
 \end{aligned} \quad (7)$$

Thus, the equivalent weight is independent of s , but the equivalent gradient scales linearly with it. As illustrated in Fig. 3, a small scaling factor ($s = 2$) results in slow convergence, whereas increasing s accelerates optimization. However, this trend does not extend indefinitely. Beyond a certain point, overly large s values amplify the gradient too strongly, leading to unstable updates and degraded generalization. In practice, we observe that moderate scaling (e.g., $s \in [4, 16]$) achieves the best trade-off between convergence speed and stability.

When examining different ranks, we observe that with a small rank (e.g., $r = 1$), the gradient norm diminishes, leading to a notable performance gap relative to e.g., $r = 64$. However, employing a higher scaling factor ($s = 16$) restores the gradient norm and narrows the gap. This effect is particularly advantageous in MoE configurations, where rank reduction across experts can be compensated by larger scaling factors, consistent with Tian et al. [68].

4. Methodology

4.1. LoRA-MoE Architecture

LoRA-MoE introduces N LoRA experts and employs a router to dynamically select and combine their outputs, mitigating task interference by isolating task-specific adaptations while maintaining parameter efficiency. Let the parameters of the i -th LoRA expert be denoted by $\{A_i, B_i\}$, and let R denote the router. Here, $B_i \in \mathbb{R}^{m \times d}$ and $A_i \in \mathbb{R}^{d \times n}$, where $d = \frac{r}{N}$. The output of the MoE system can then be expressed as:

$$y = W^{(0)}x + \sum_{i=1}^N R(\mathbf{x})_i (sB_i A_i \mathbf{x}), \quad (8)$$

where $R(\mathbf{x})_i$ is the router score for the i -th expert, $W^{(0)}$ is the pre-trained weight matrix, and s is the LoRA scaling factor.

In dense gating, the router consists of a dense layer with trainable parameters W_g . The gating scores are computed using a softmax function:

$$R(\mathbf{x})_i = \text{softmax}(W_g^\top \mathbf{x}). \quad (9)$$

Dense gating is employed in the soft routing strategy, where the outputs of multiple experts are weighted and combined according to their gating scores.

We adopt the Mixtral top- k router [31]. Specifically, an MoE layer comprises N linear modules $\{W_1, \dots, W_N\}$ and a router $R_z \in \mathbb{R}^{m \times N}$ that assigns input $\mathbf{x}$ to experts based on routing scores:

$$p^i(\mathbf{x}) = \frac{\exp(z^i(\mathbf{x}))}{\sum_{j=1}^N \exp(z^j(\mathbf{x}))}, \quad (10)$$

where $z(\mathbf{x}) = W_z \mathbf{x}$ and $p^i(\mathbf{x})$ is the score for expert i . Let $S_k(x) \subset \{1, \dots, N\}$ denote the indices of the top- k experts according to their gating logits $z(x)$, ensuring $|S_k(x)| = k$ and $z_i(x) > z_j(x)$ for all $i \in S_k(x)$ and $j \notin S_k(x)$. The normalized top- k gating weights are defined as:

$$R(\mathbf{x})_i = \begin{cases} \frac{\exp(z_i(x))}{\sum_{j \in S_k(x)} \exp(z_j(x))}, & i \in S_k(x), \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

The final LoRA MoE output combines the frozen base layer with the weighted contributions of the selected LoRA experts:

$$\text{MoE}_{\text{LoRA}}(\mathbf{x}) = W^{(0)}x + \sum_{i \in S_k(x)} R(\mathbf{x})_i (sB_i A_i \mathbf{x}). \quad (12)$$

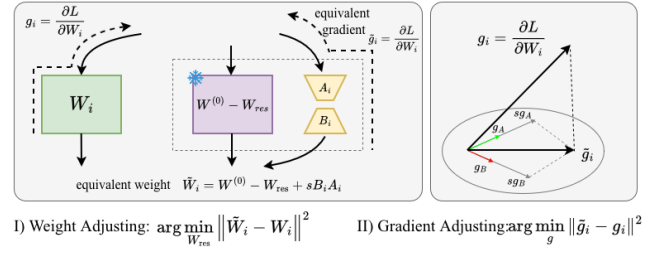


Figure 4. Optimization with SVD-structured MoE by separately aligning each expert. W_{res} ensures the equivalent weight equals $W^{(0)}$ before optimization. Scaling aligns each expert’s equivalent gradients to those of full MoE FT.

During training, only the experts indexed by $S_k(x)$ and their corresponding gating paths receive gradient updates. Since $k \ll N$, LoRA-MoE activates significantly fewer parameters than a dense MoE.

4.2. Adaptive Priors Initialization

Building on the relevance of different SVD components on input data (Sec. 3.1), we introduce a novel initialization methodology for a LoRA-MoE framework. This strategy assigns distinct, non-overlapping segments of a pre-trained model’s SVD to individual experts. The core hypothesis is that *the MoE’s routing mechanism can learn to dynamically select the expert whose assigned singular value spectrum is most pertinent to a given input, creating a more adaptive low-rank approximation.*

Define a set of SVD component segments, $\mathcal{E}_r$, which are distributed evenly among the N available experts:

$$\mathcal{E}_r = \left\{ (U_{[t, k:k+d]}, S_{[k:k+d, k:k+d]}, V_{[k:k+d, :]}^\top) \mid j = 1, \dots, N \right\}, \quad (13)$$

where $t = \frac{\min(m, n)}{N}$ and the starting index for the j -th expert is $k = (j - 1)t$. Each expert is allocated a uniform rank of $d = \frac{r}{N}$. Subsequently, each expert’s constituent low-rank matrices, B_i and A_i , are formed from a unique SVD segment $(U', S', V'^\top) \in \mathcal{E}_r$ as follows:

$$B_i = \sqrt{\frac{1}{s}} U' S'^{1/2} \in \mathbb{R}^{m \times d}, \quad A_i = \sqrt{\frac{1}{s}} S'^{1/2} V'^\top \in \mathbb{R}^{d \times n}. \quad (14)$$

The scaling factor $\sqrt{s}$ is applied to ensure that the resulting update matrix sBA is invariant to s [57]. This structured initialization promotes expert specialization and enhances the model’s flexibility for diverse FT tasks.

4.3. Theoretical Optimization Alignment

The direct integration of SVD spectral subspaces into MoE architectures introduces weight misalignment and complex gradient dynamics. These issues are not present in conventional LoRA that employs zero-initialization. This specific problem within MoE frameworks has been largely underexplored. We address this gap by first formally stating the necessary alignment requirements for stable training, and subsequently demonstrating that these requirements can be satisfied through a principled scaling mechanism.

Theorem 1 (Single model alignment.). *Let $\tilde{W}_t$ denote the effective LoRA weight and $\tilde{g}_t$ its effective gradient, with W_t and g_t the corresponding quantities for full FT. Alignment follows from the pair of conditions:*

$$\tilde{W}^{(0)} \approx W^{(0)}, \quad \tilde{g}_t \approx g_t, \quad (15)$$

(defined in Eq. 4 as the “equivalent weight” and “equivalent gradient”). These conditions mitigate the performance gap in single-adapter LoRA [77, 80].

Theorem 2 (MoE alignment.). *For an MoE with experts $i \in \{1, \dots, N\}$, router weights $R(\mathbf{x})_i$ (top- k gating), and per-expert LoRA parameters, alignment reduces to enforcing*

$$\tilde{W}_i^{(0)} \approx W_i^{(0)}, \quad \tilde{g}_i^t \approx g_i^t, \quad \forall i \in \{1, \dots, N\}. \quad (16)$$

This makes our LoRA-MoE training dynamics equivalent to those of an upcycled MoE [19] with full-rank FT. This equivalence is highly desirable as it permits the optimization of each expert to proceed independently, simplifying the overall learning process and enhancing stability.

Initialization alignment. We enforce this equivalence at initialization by adopting the principles of upcycled MoE [19]. The procedure begins with the pre-trained base weight $W^{(0)}$. For each expert, we initialize its low-rank factors $B_i^{(0)} A_i^{(0)}$ using SVD priors. The resulting effective weight for the LoRA-MoE architecture is thus defined as:

$$\tilde{W}^{(0)} = W^{(0)} - W_{\text{res}} + \sum_{i=1}^N R(\mathbf{x})_i s B_i^{(0)} A_i^{(0)} \approx W^{(0)}, \quad (17)$$

To ensure that the equivalent weight $\tilde{W}^{(0)}$ aligns with the pre-trained weight $W^{(0)}$ at initialization (see Eq. 15), we introduce a residual compensation term W_{res} defined as the expected weighted contribution of all LoRA experts:

$$W_{\text{res}} = s \mathbb{E}_{\mathbf{x}} \left[\sum_{i=1}^N R(\mathbf{x})_i B_i^{(0)} A_i^{(0)} \right]. \quad (18)$$

This term preserves the unbiased initialization of $\tilde{W}_0$.

Theorem 3 (Router moments.). *For top- k gating over N experts with input data denoted by $\mathbf{x}$, the router’s behavior is simplified through the use of moment identities:*

$$\mathbb{E}_{\mathbf{x}}[R(\mathbf{x})_i] = \frac{1}{N}, \quad \text{Var}(R(\mathbf{x})_i) = \frac{N-k}{kN^2}, \quad (19)$$

for all $i \neq j \in \{1, \dots, N\}$. The variance of the initialization mismatch term, $W_{\text{res}} - s \sum_{i=1}^N R(\mathbf{x})_i B_i^{(0)} A_i^{(0)}$, is proportional to the aggregate $\sum_{i=1}^N B_i^{(0)} A_i^{(0)}$. Given that in sparse gating scenarios (e.g., $2k < N$), the expert weight standard deviation, $\text{std}(R(\mathbf{x})_i)$, constitute an important part of the mean $\mathbb{E}[R(\mathbf{x})_i]$, the effective control of this variance becomes critical.

Theorem 4 (Residual matching objective.). *To mitigate the initialization mismatch, we determine the optimal constant residual weight W_{res} by minimizing the expected MSE:*

$$W_{\text{res}}^+ = \arg \min_{W_{\text{res}}} \mathbb{E}_{\mathbf{x}} \left\| W_{\text{res}} - s \sum_{i=1}^N R(\mathbf{x})_i B_i^{(0)} A_i^{(0)} \right\|^2. \quad (20)$$

By substituting the established router moments from Eq. 19, we derive the following closed-form solution:

$$W_{\text{res}}^+ = \frac{s}{N} \sum_{i=1}^N B_i^{(0)} A_i^{(0)}. \quad (21)$$

Note that contemporary zero-initialization schemes for LoRA-MoE [68, 90] are recovered as the special case where W_{res}^+ vanishes.

SVD-consistent scaling. To simultaneously maintain the information embedded in the SVD spectral subspaces in the pre-trained model and enforce numerical stability (i.e., by ensuring a tighter approximation in Eq. 17), we damp per-expert low-rank initial factors by $\rho > 0$:

$$B_i^{(0)} = \sqrt{\frac{1}{s\rho}} U_i S_i^{1/2}, \quad A_i^{(0)} = \sqrt{\frac{1}{s\rho}} S_i^{1/2} V_i^\top. \quad (22)$$

This joint scaling strategy effectively reduces the variance in initialization mismatch.

Gradient alignment. The theoretical analysis of the gradient dynamics under zero-initialized LoRA (where $B^{(0)} = 0$, $A^{(0)} \sim U(-\sqrt{6/n}, \sqrt{6/n})$) reveals that the effective LoRA gradient $\tilde{g}_i^t$ for expert i is approximated by:

$$\tilde{g}_i^t = s^2 \left(B_i^t B_i^{t\top} g_i^t + g_i^t A_i^{t\top} A_i^t \right). \quad (23)$$

Theorem 5. *The optimal scaling factor s^* , obtained by minimizing $\|\tilde{g}_i^t - g_i^t\|$ under a learning-rate ratio η (Full FT vs. LoRA), is given by:*

$$s^* = \sqrt{\frac{3n\eta}{r}}. \quad (24)$$

Given the common practical condition $n \gg r$, the optimal factor s^* is notably greater than two, providing a theoretical justification for why small scaling factors are insufficient and validating the necessity of moderate scaling (see Sec. 3.2).

Practical notes on SVD spectral subspaces. While the exact gradient dynamics for SVD-subspaces from the pre-trained model are hard to analyze, increasing s and ρ in Eq. 22 attenuates the initial magnitude of $B_i^{(0)}$ and $A_i^{(0)}$, pushing the system towards the zero-initialization regime where Eq. 24 is valid. We adopt this scaling in MedQwen.

5. Experiments

We evaluate MedQwen along three axes: (1) accuracy gain on medical VQA compared to Med-VLM baselines, (2) clinical report generation on two distinct benchmarks, and (3) robustness across different training configurations and data regimes.

5.1. Experimental Setups

Implementation Details. We select Qwen2.5-VL 7B as the base model for our architecture (Fig. 2). MedQwen is initialized with the pre-trained weights of Qwen2.5 (7B). To ensure a fair comparison with prior work, we closely follow the configurations used in previous studies [39, 44]. Details of the baseline setups are provided in Appendix 8.2.

Table 1. Comparison of MedQwen with other VLMs and unified multi-modal models on medical visual comprehension tasks. **Bold** and underlined text indicates the best performance and second-best performance, respectively. Close-ended: yes/no and other limited choices. Open-ended: Do not have a limited question structure and could have multiple correct answers.

Model	# Params	Medical VLM	VQA-RAD $\uparrow$		SLAKE $\uparrow$		PathVQA $\uparrow$		OMVQA $\uparrow$	Avg. $\uparrow$
			close	open	close	open	close	open		
BLIP-2	6.7B	X	43.4	20.5	41.6	32.1	48.5	23.8	26.9	33.8
LLaVA-v1.5	7B	X	51.8	26.5	37.1	29.8	53.5	26.7	44.7	38.6
InstructBLIP	7B	X	61.0	28.2	66.8	40.7	56.0	27.3	29.0	44.1
Yi-VL	6B	X	52.6	27.1	52.4	30.8	54.9	25.7	50.2	41.9
InternVL2	8B	X	64.9	33.8	66.6	35.2	60.0	34.6	54.5	49.9
Llama-3.2	11B	X	68.9	29.3	<u>72.4</u>	37.1	62.8	36.5	63.2	52.9
Qwen-2.5-VL	7B	X	61.8	27.2	64.7	36.7	60.5	33.4	60.8	49.3
LLaVA-Med	7B	✓	68.9	32.5	57.7	41.3	52.5	30.3	31.8	45.0
Med-Flamingo	8.3B	✓	67.4	27.0	46.4	23.8	51.3	29.5	24.6	38.6
HuatuoGPT-Vision	7B	✓	71.1	37.7	58.5	45.6	52.9	30.8	39.9	48.1
HealthGPT-M3	3.8B	✓	69.6	37.2	56.4	43.6	50.1	27.7	32.5	45.3
HealthGPT-L14	14B	✓	<u>74.5</u>	<u>54.5</u>	71.9	<u>56.2</u>	<u>75.2</u>	<u>42.1</u>	<u>67.2</u>	<u>63.1</u>
MedQwen	7B	✓	78.8	59.6	75.3	59.9	84.2	49.1	70.6	68.2

Table 2. Accuracy (%) of different Med-VLMs based on LLaVA-Med-1.5 on medical VQA task.

Models	Radiology		Ophthalmology		Pathology	
	IU-Xray	MIMIC-CXR	Harvard-FairVLMed	Quilt-1M	PMC-OA	
GPT-4o	63.25	60.61	61.50	53.56	49.70	
Gemini-1.5	59.73	61.02	58.53	56.88	52.17	
LLaVA-v1.6	58.05	63.70	48.52	35.73	38.54	
Qwen-2.5-VL	59.43	60.43	38.06	28.74	29.53	
InternVL-2	54.06	59.47	44.38	37.82	34.40	
LLaVA-Med-1.5	<u>75.47</u>	<u>75.79</u>	63.03	<u>62.80</u>	<u>59.28</u>	
Med-Flamingo	26.74	61.27	42.06	27.11	32.62	
MedVInT	73.34	66.06	35.92	26.81	27.77	
RadFM	26.67	69.30	52.47	27.02	25.12	
miniGPT-Med	54.87	53.92	<u>66.73</u>	26.82	27.03	
MedQwen	90.33	84.68	88.43	73.74	66.32	

Table 3. Model performance (%) of different methods on report generation task.

Metric		RadFM	LLaVA Med	Med Flamingo	InternVL	MedDr	MedQwen
MIMIC-CXR	F1-RadGraph	18.25	6.36	7.13	6.70	<u>22.44</u>	29.40
	BLEU-1	22.19	19.34	22.43	25.58	<u>27.28</u>	33.74
	BLEU-4	5.55	0.96	1.91	1.69	7.59	<u>7.38</u>
	ROUGE-1	28.88	21.44	21.69	22.87	<u>32.58</u>	36.43
	ROUGE-L	20.52	13.90	14.60	15.61	<u>22.59</u>	25.69
	CheXbert Vec	31.18	15.56	18.69	15.90	<u>34.18</u>	44.19
	METEOR	20.42	13.25	14.02	17.54	<u>23.77</u>	25.47
IU-Xray	F1-RadGraph	29.17	4.72	11.91	11.21	<u>33.19</u>	37.76
	BLEU-1	<u>40.78</u>	14.85	16.28	18.27	37.71	41.59
	BLEU-4	10.28	0.68	2.06	1.73	12.22	<u>11.61</u>
	ROUGE-1	36.79	12.98	14.93	18.22	<u>39.49</u>	42.06
	ROUGE-L	26.07	9.96	11.11	13.43	<u>28.35</u>	29.65
	CheXbert Vec	<u>59.26</u>	14.01	24.25	24.17	56.47	64.93
	METEOR	30.92	14.48	17.69	20.03	<u>32.34</u>	33.38

Datasets and Training Stages. Our model undergoes a three-stage training process. During the first and second phases, we utilize well-structured datasets provided by LLaVA-Med [39] for *alignment* and *instruction tuning*. In the third phase, we employ medical datasets for *MoE-tuning*, including SLAKE [47], VQA-RAD [37], and PathVQA [20], among others, covering a wide range of modalities and anatomical regions. De-

tails of the datasets and their corresponding modalities are provided in Table S1 and Table S2.

5.2. Methods on Med-VQA

On medical VQA, we compare MedQwen with both medical-specific and general-purpose LVLMs (Tables 1 and 2). As shown in Table 1, MedQwen achieves superior performance across nearly all metrics and datasets, surpassing the Qwen-2.5-VL baseline by an average of 18.9% and outperforming Med-LLaVA and Med-Flamingo by 23.2% and 29.6%, respectively. To evaluate generalization, we also compare MedQwen with LVLMs trained on large-scale open-domain data (Table 2); MedQwen consistently outperforms these models, confirming its strong generalizability across diverse image domains and medical multimodal tasks.

5.3. Medical report generation

We also investigate the evaluation of LVLMs for the medical report generation task (Table 3). In MIMIC-CXR, our method significantly outperforms LLaVA-Med and MedDr methods, achieving improvements of up to 23% and 6.9% in F1-RadGraph, and 28.6% and 10% in CheXbert, respectively. Furthermore, on IU-Xray, MedQwen surpasses RadFM, a model primarily focused on radiology tasks, across nearly all evaluated metrics. For instance, MedQwen achieves a BLEU-1 score of 41.59 and a CheXbert score of 64.93 on the IU-Xray dataset, underscoring its proficiency in comprehensively interpreting medical images. These empirical results demonstrate that the MoE effectively specializes to diverse types of medical data while mitigating inter-dataset conflicts.

5.4. Zero-Shot Classification

Table 4 reports zero-shot accuracies across nine radiology benchmarks spanning four imaging modalities. Our method achieves 95.31% of the full FT MoE performance (58.8% vs

Table 4. Zero-shot classification evaluation of BiomedCLIP ViT-B/16 with full fine-tuning and LoRA variants (rank is 8 if not specified) across various radiology datasets.

Approach	# Params (%)	X-ray		Ultrasound		MRI		CT			Avg.
		CheXpert(5x200)	RSNA	Thyroid	Breast	ACL	Meniscus	Axial	Coronal	Sagittal	
Full FT	100	65.30	72.56	76.05	76.12	85.89	40.09	40.13	28.09	26.59	56.76
Full FT MoE	760	72.39	74.53	78.59	88.50	87.38	46.34	45.09	33.40	29.27	61.72
<i>Single LoRA Methods</i>											
LoRA	1.49	54.35	69.81	62.72	72.13	79.03	33.89	35.23	27.41	21.69	50.70
LoRA (rank16)	2.99	61.07	72.37	63.28	74.98	82.71	37.94	42.79	30.45	26.84	54.71
LoRA (rank32)	5.98	61.88	73.17	64.13	75.87	83.65	39.55	42.85	30.55	27.44	55.45
MiLoRA	1.49	52.68	69.19	61.52	71.92	78.38	33.68	34.13	26.07	20.77	49.82
PiSSA	1.49	53.41	68.62	63.11	71.48	77.52	32.92	34.49	27.68	21.50	50.08
<i>LoRA MoE Methods</i>											
MoELoRA	2.24	62.36	72.80	64.85	74.89	83.83	40.53	41.91	30.84	27.65	55.52
AdaMoLE	2.33	60.10	72.35	63.17	75.38	82.69	39.15	41.70	30.42	26.51	54.61
HydraLoRA	1.58	61.77	71.56	64.24	75.08	82.51	38.73	38.26	29.55	24.20	53.99
Ours	2.24	66.83	74.14	77.88	77.23	86.57	42.45	44.05	32.10	28.24	58.83

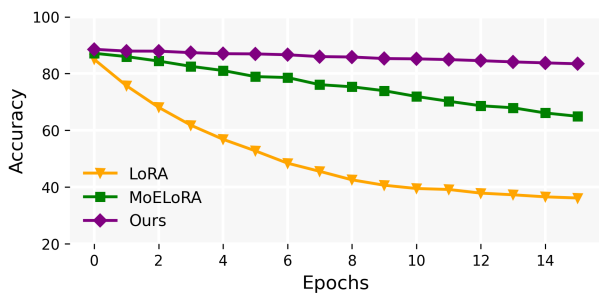


Figure 5. Comparison of catastrophic forgetting across different fine-tuning methods in sequential learning. The plot illustrates the accuracy decay over 15 epochs for LoRA, MoELoRA, and our method.

61.7%) despite using 339 times fewer parameters (2.24 vs. 760). It also outperforms the strongest single-LoRA baseline (rank-32, 55.45%) by +3.38 points with 2.7 times fewer parameters (2.24% vs. 5.98%), and improves over PiSSA by 8.75% and HydraLoRA by 4.84%. Among LoRA-based methods (single and MoE), our approach achieves the best result on *all nine* datasets and further outperforms full FT (56.76%).

Table 5. Ablation study of MedQwen. MoE denotes using the MoE architecture instead of a single LoRA. MS refers to using MoE scaling. O, P, M, and R represent different SVD-based initializations: original MedQwen segments, segments containing the principal singular values, segments containing the minor singular values, and randomly selected segments, respectively.

MoE	SVD Initialization				Avg.	Avg. (w/o MS)
	P	M	R	O		
✓					67.3	66.8
✓	✓				67.4	66.6
✓		✓			67.6	66.9
✓			✓		67.7	66.9
				✓	/	60.1
✓				✓	68.2	67.3

5.5. Catastrophic Forgetting

VLMs face a fundamental challenge when exposed to sequential or non-stationary learning scenarios—catastrophic forgetting [35, 82, 91]. This phenomenon refers to the abrupt and severe loss of previously acquired knowledge when a model is

trained on new data or tasks. In other words, during sequential learning, new information tends to overwrite the representations formed from earlier experiences, leading to a substantial degradation in performance on previously learned tasks.

To examine the vulnerability of existing fine-tuning methods, such as LoRA and MoELoRA, to catastrophic forgetting during instruction tuning, we designed a simple yet effective evaluation framework. Specifically, we first fine-tuned the Qwen model using three different methods including our proposed MoE-SVD, standard LoRA, and MoELoRA on the Harvard-FairVLMed dataset. Subsequently, we fine-tuned these models on the PathVQA dataset, which belongs to a completely different medical domain, and evaluated their performance on the original Harvard-FairVLMed dataset to measure the degree of forgetting associated with each method. As shown in Fig. 5, the results reveal that the LoRA-based model suffered a drastic performance drop of over 50%, while MoELoRA exhibited a moderate reduction of more than 20%. In contrast, our proposed MoE-SVD approach demonstrated remarkable resistance to catastrophic forgetting, with only a 5% decrease in accuracy after 15 epochs of sequential training. This finding underscores the effectiveness of our proposed method in preserving prior knowledge while maintaining adaptability across diverse domains. Detailed analyses are in Appendix 9.2.

5.6. Hallucination Evaluation

We further include a medical hallucination evaluation in the Appendix 10, which shows that MedQwen consistently outperforms both general-purpose and medical-specific LVLMS.

5.7. Ablation Studies

We ablate the impact of our adaptive priors initialization and gradient scaling (Table 5). Our initialization, with or without MoE scaling, consistently outperforms other methods (note that no SVD initialization corresponds to the original zero initialization, yielding 67.3/66.8). Without MoE, initializing a single LoRA with our SVD fragments achieves a performance of 60.1. In contrast, our MoE architecture achieves 67.3, demonstrating the advantages of expert specialization

Table 6. Comparison of our method against MoELoRA and HydraLoRA in memory cost, training time, and performance. Memory cost was measured and training time was recorded on the medical VQA benchmarks using one A100 GPU with identical batch sizes.

Model		VQA-RAD		SLAKE		PathVQA		OMVQA	Params(%)	Memory Usage	Epoch Time
		close	open	close	open	close	open				
MedQwen w/	+MoELoRA	72.3	57.2	70.5	53.7	78.4	38.6	65.1	0.96	32.56 GB	29h15min
	+HydraLoRA	73.6	57.4	66.4	52.4	74.2	36.0	64.9	0.84	32.47 GB	29h17min
	+Ours	78.8	59.6	75.3	59.9	84.2	49.1	70.6	0.96	32.56 GB	29h19min

and the alignment strategy.

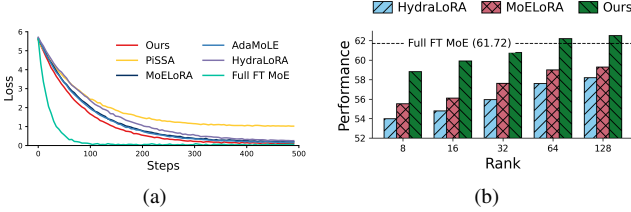


Figure 6. (a) Training loss curves of different LoRA methods and Full Fine-tuning MoE on radiology datasets. The balance loss is excluded in the MoE baselines for a fair comparison with single LoRA baselines. (b) Performance of different methods across ranks.

5.8. Convergence Speed

As shown in Fig. 6a, we compare the training loss curves of PiSSA, various LoRA MoE baselines, our proposed method, and Full FT MoE on radiology datasets. Our method demonstrates faster convergence than all LoRA MoE baselines and achieves performance closest to Full FT MoE. Notably, our method achieves a lower final loss, balancing performance and efficiency. In contrast, methods like PiSSA exhibit rapid initial drops but plateau at higher final loss (see Sec. 3.1).

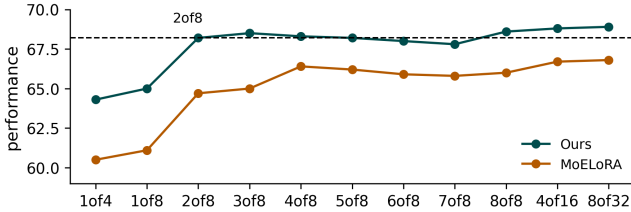


Figure 7. Performance vs. number of experts and activation ratio (total rank=32).

5.9. Scalability

Scaling with Rank. To evaluate the scalability, we increase the rank in MedQwen from 8 to 128 on radiology datasets (Fig. 6b). As the rank increases, the performance gap between MedQwen and full fine-tuning MoE narrows significantly. Notably, MedQwen consistently outperforms both MoELoRA and HydraLoRA across all ranks. At rank 32, MedQwen achieves 60.78, surpassing MoELoRA (57.63) by 3.15% and HydraLoRA (55.94) by 4.84%. While higher ranks improve performance, gains diminish. For instance, MedQwen improves by 0.3% from rank 64 to 128, indicating limited benefit despite higher computational costs.

Scaling with Expert Count and Activated Ratios. We also conduct experiments on medical VQA datasets with the total rank fixed at 32 (Fig. 7). Key findings include: (1)

With 8 experts, the 2-of-8 configuration achieves strong performance. Activating more experts may yield lower performance, showing that sparse expert activation is important. (2) Increasing the total number of experts may improve performance, as seen in 2-of-8 vs. 4-of-16 / 8-of-32, but makes routers harder to train, increases memory consumption, and reduces runtime efficiency. (3) Our method consistently outperforms MoELoRA, especially when activating only one expert, consistent with discussions in Sec. 3.2. In practice, 2-of-8 offers a balanced trade-off between performance and storage efficiency.

5.10. Computation Analysis

Parameter Size. The Params (%) column (Table 6) compares parameter ratios of LoRA baselines and MedQwen. Our model achieves state-of-the-art performance with a parameter size of $O(Hr) + O(He)$, much smaller than Full FT’s $O(H^2)$ and Full FT MoE’s $O(kH^2)$. Since $r, e \ll H$, MedQwen is much more efficient. See Appendix 11 for details.

FLOPs and Runtime We estimate the memory usage and performance of FT MoE based on single-GPU runtimes. As shown in Table 6, the LoRA-MoE series trains much faster than Full FT MoE. Among LoRA-MoE variants, our method achieves the best performance with identical memory and time costs. FLOPs analysis (Sec. 11.1) reveals that Full FT MoE scales as $O(ksH^2)$, while LoRA MoE simplifies to $O(sH^2)$ since $k < e$ and $r \ll H$. Thus, LoRA MoE’s FLOPs remain nearly constant, independent of k , unlike Full FT MoE, which scales linearly with k .

6. Conclusion

We propose MedQwen, a novel large medical vision-language model that unifies comprehension and generation through an enhanced LoRA fine-tuning approach. MedQwen adaptively integrates SVD-structured priors and aligns low-rank gradients with the fully fine-tuned MoE via theoretical scaling. Without modifying the underlying architecture or training algorithms, MedQwen substantially improves both efficiency and performance, achieving state-of-the-art results across 23 diverse medical comprehension and generation tasks. These results demonstrate its strong potential for real-world healthcare applications.

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